



Jeroen Bouma

Quantitative Investment Strategist

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github.com/JerBouma [A2](#) Dutch and English (professional)

PROFILE

My background is in Quantitative Finance and most of my work comes down to putting investment theory into practice with Python. I am currently a Quantitative Investment Strategist at a.s.r. asset management, where I help build, maintain and use the models for Solvency II internal model calculations, lifecycle investing, portfolio optimization and ALM across multi-billion euro portfolios. I also contribute to our work at the intersection of quantitative finance and AI, from forecasting models to generative AI built on agentic frameworks and retrieval-augmented generation (RAG).

Outside of my day job, I've built open-source Python libraries such as the Finance Toolkit and Finance Database. They are used by thousands of analysts, developers and students worldwide.

EXPERTISE

Quantitative Finance	• Strategic Asset Allocation	• Solvency II Internal Models
	• Asset Liability Management	• Economic Scenario Generation
	• Lifecycle Investing	• Econometrics & Time Series
	• Portfolio Optimization	• Factor Models
	• Liability Hedging	
Artificial Intelligence	• Agentic Frameworks	• Machine Learning
	• RAG Pipelines	• AI Evaluation & Guardrails

EXPERIENCE

Quantitative Investment Strategist February 2024 – Present a.s.r. asset management

As part of the team I have contributed significantly to both building and using the models behind our Solvency II internal model calculations, lifecycle investing and portfolio optimization, as well as to forecasting and generative AI applications built on agentic frameworks and RAG that help us analyze portfolios and risk. I also contribute to the yearly SAA through the economic scenario calibrations and guide several working students in their day-to-day work and with their theses.

Financial Risk Analyst June 2023 – February 2024 a.s.r. asset management

I analyzed profitability and investment risk under different risk frameworks (Solvency II and internal models), looked at portfolio sensitivities and how they relate to the liabilities, and assessed the impact of month-to-month changes. I also helped develop Python models to improve valuation and financial risk measurement.

EDUCATION

Register Beleggingsanalist (RBA)

CFA Society Netherlands
September 2020 – June 2023

MSc Banking & Finance

Utrecht University
September 2017 – June 2019

Bachelor of Economics (BSc)

Hanze University of Applied Sciences
September 2013 – July 2017

CERTIFICATIONS

Transforming Your Business with AI INSEAD

June 2026
[Verify credential](#)

Certified ScrumMaster (CSM) Product Owner

August 2022

Chartered Financial Analyst (CFA)

Level 1
January 2020

SKILLS

Python Machine Learning
Agentic Frameworks RAG
GLASS SQL Databricks
Azure PowerBI Git
JavaScript Jira
Confluence MATLAB VBA
MCP Docker GitHub R

Product Manager

February 2022 – June 2023

OpenBB

OpenBB set out to compete with Bloomberg through open source. As Product Manager I led a multi-disciplinary team of engineers and finance professionals using Scrum, and focused on translating financial theory into code, building features such as the economy, portfolio and fixed income menus myself. Towards the end I also led product marketing and the outreach to universities.

ALM Advisor

July 2019 – February 2022

PGGM

I gave actuarial and strategic advice to several Dutch pension funds. I was responsible for developing and maintaining the econometric model in MATLAB and led most of the technical development of the new econometric model in Python. These models performed the ALM, URM and VEV calculations. I also worked on interactive dashboards and GUIs and on applying Machine Learning techniques.

OPEN SOURCE

Finance Database

2020 – Present

Creator & Maintainer

 github.com/JerBouma/FinanceDatabase

★ 9,300+ GitHub stars ⬇ 600,000+ downloads 🍴 950+ forks

An open-source database with 300,000+ symbols covering Equities, ETFs, Funds, Indices, Currencies, Cryptocurrencies and Money Markets. It gives an overview of sectors, industries and investment types across 111 countries. I started it out of frustration that I did not know which financial products existed. OpenBB later came across the project, which led to a job offer.

Finance Toolkit

May 2019 – Present

Creator & Maintainer

 github.com/JerBouma/FinanceToolkit

★ 5,300+ GitHub stars ⬇ 640,000+ downloads 🍴 600+ forks

An open-source toolkit with 500+ financial methods, each written to be as transparent as possible. It supports Equities, Options, Currencies, Cryptocurrencies, ETFs, Mutual Funds, Indices, Commodities, Key Economic Indicators and more. Combined with the Finance Database, it lets you do a full competitive analysis. It is used by thousands of analysts, developers and students worldwide.

HOBBIES

Tinkering

I like to tinker with technology at home, from home automation to mini PCs and home servers. Lately that includes experimenting with LLMs and agentic setups, next to the many programming experiments I run in the financial domain.

Food & Wine

I love cooking and wine tasting, with a real adoration for champagne.

Sports

I stay active through running, swimming and cycling. Combining the three into a triathlon is on my list for some day.

REFERENCES



Didier Lopes

CEO at OpenBB

"Jeroen was at the intersection of finance, programming and open source - which is a combination that is very hard to find ... ultimately, the team saw Jeroen as the go-to person when financial theory was involved. ... I highly recommend Jeroen."



Luuk van Benthem

Senior ALM Adviseur at PGGM

"In the implementation phase Jeroen played a key role on the development of the technical side of the model. Meanwhile he did a good job at educating our team on these new topics and providing good documentation. ... Over time I learned a lot from Jeroen which I'm very thankful for!"



Pascal Janssen

Strategisch ALM Adviseur at PGGM

"Jeroen is highly skilled in every aspect of programming in Python. His knowledge and experience with Python have added great value to my department. ... Jeroen works in a very structured way and approaches his own development with a positive attitude."

(translated from Dutch)

 More about my referees and their full recommendations on LinkedIn